



DISPLAY REFERENCE

Matrix Display Guide

Recognise animations, mode letters, pairing, and startup feedback.

2026 EDITION / IAIN BENNETT / MIT LICENSE

The blue LED matrix on the **VirtualGlove Controller (Arduino UNO Q)** is VirtualGlove's status display. It tells you which mode is active and helps distinguish startup, practice, tracking, and pairing. It does not show a game's score or confirm that a game accepted a button.

Use the pictures and quick checks below to recognise what is happening and decide what to do next.

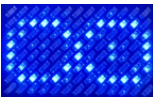
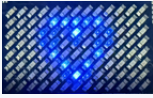
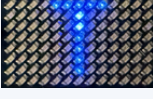
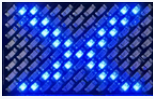
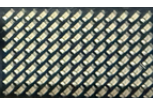
Start here

After reboot, the display shows Router's neutral animation. Start a registered game and the display changes to that app's cues automatically. When the game ends, it returns to neutral.

You can also open the controller address to choose an app while no game is running. Finish a game before pairing or changing apps.

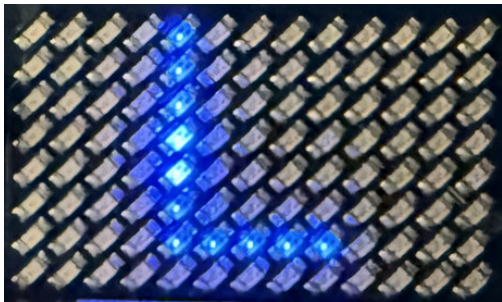
If the display stays neutral during your game, open Dashboard and check the console connection and selected game. You do not need to reinstall just because no game is selected.

Recognise the display

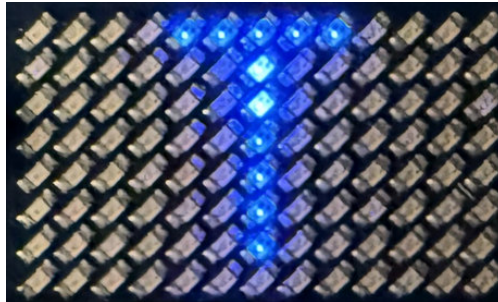
What you see	See it	What it means	What to do
Arduino boot logo		The VirtualGlove Controller's system software is starting, before VirtualGlove controls the display.	Wait for the app's hourglass or normal display.
System heart animation		The board is progressing through system startup.	Wait for the app display.
Router's neutral animation	-	No selected product has an active Matrix request.	Start a registered game or open the controller home page to choose an app.
Pulsing hourglass		VirtualGlove is starting.	Allow startup to finish. If it persists, check Dashboard.
A large scanning L		Play or Glove Academy lessons are active. L stands for local play or lessons.	Follow the game or practice moves shown in your browser; controller output is paused.
A large scanning T		Gesture tuning is active, including hand setup.	Follow the recording, preview, and save instructions in Glove Academy. Controller output is paused.
A steady 1-14 , A-I , BS , or GB		A game profile is selected, but a calibrated hand is not currently being reported as tracked.	Show your hand and check tracking/calibration on Dashboard. Program 14 intentionally keeps the camera off.
1-13 , A-I , BS , or GB gently changing brightness		The app reports a detected, calibrated hand for that profile.	Check Dashboard's controller status before playing. This pulse alone does not mean controls are enabled.
ID , characters, PN , and digits repeating	-	Secure pairing is showing the device identity and temporary PIN.	Follow the Setup page; read each group in order.
A flashing X		The app has requested an error display.	Read Dashboard's error message before deciding whether to reconnect the camera or restart.
A blank matrix		The display has been turned off, the app is stopping, or the board is still starting. It may also have lost power.	Use the browser and board power indicators to distinguish these cases. Blank does not prove shutdown is complete.

Local play and Glove Academy: L and T

L: Local play or Glove Academy lessons



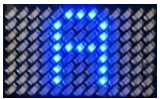
T: Gesture tuning



L means local Play or Glove Academy lessons are active. **T** means gesture tuning is active, including **Set up my hand**, individual adjustments, and previews. Follow the browser's instructions to practice or complete your recordings.

Both modes pause controller output. After practice, use Dashboard to check the selected profile and controller state; after tuning, explicitly start controller delivery when ready to play. A pairing display can temporarily cover either letter. Tracking details and pose failures remain in the browser rather than being spelled out on the matrix.

Game profile codes

Code	See it	Selected profile
1 through 13	Numeric program code	The corresponding original camera-active numeric profile; older firmware safely shows blank
14	Numeric program code	Program 14 is selected; camera and VirtualGlove output intentionally stay off while the game session remains visible
A through I		The corresponding reusable Program A-I profile; A is shown
BS		Bad Street Brawler
GB		Super Glove Ball

For example, **GB** is what you should expect with Super Glove Ball selected. It stays steady until a calibrated hand is being tracked, then gently pulses. The characters remain the same: the animation does not identify individual finger curls, movements, or button presses.

The important distinction is **tracking versus delivery**. A pulsing **GB** can appear while controller output is stopped. Confirm **Start controller** has been used and Dashboard shows output enabled, then confirm the action in the game. The matrix does not acknowledge console receipt or the game's response.

A generic **PG** ready symbol or a small pulsing tracking symbol can appear when no recognised profile identifier is available. These are fallback displays; normal named game profiles use their numeric, letter, or dedicated two-character

codes. They do not represent extra games or new gesture commands. If you expected **GB** or **BS**, check the selected profile on Dashboard.

Pairing: ID and PN

When a submitted pairing attempt finishes, the matrix releases the approval PIN and resumes its normal display. The selected app resumes its display. With no app selected, Controller Router shows its neutral animation. VirtualGlove's On, Dim, or Off attract setting applies while VirtualGlove is selected; game and status displays take priority.

During secure pairing, the small display presents the information in pieces:

1. **ID** announces the device certificate identity.
2. Seven hexadecimal characters follow in groups of three, with the last character shown alone.
3. **PN** announces the temporary six-digit PIN.
4. Two groups of three digits follow, preserving leading zeroes.
5. The sequence repeats so you can read it again.

Read each group from left to right and join the groups in the order shown. Do not mistake **PN** for a game program or use the certificate identity as the PIN.

Follow the Setup page to compare the identifier and enter the PIN. Pairing information temporarily replaces the usual display. If it expires before you finish, prepare a new pairing attempt. Check the page for confirmation that pairing succeeded.

For public photographs, use a clearly marked demonstration sequence or obscure live pairing digits. Never publish an active PIN or connection credentials. See [secure pairing](#) for the full procedure.

Errors, pauses, and a blank display



A blinking **X** means the app needs attention. Read Dashboard's error message for the cause and the next step. Check Dashboard whenever something is not working, even if the matrix still shows a normal mode.

A blank display alone does not prove the board is safe to unplug. Use the normal Shutdown procedure and its completion guidance. If the board should be running, check its power, wait for boot to finish, and try Dashboard. If the web app works but the display stays blank, record what appeared immediately before it went blank and whether restarting the app changes it.

For older standalone display settings and firmware maintenance, see the [Configuration Reference](#).